MODULE 1: FOUNDATION		
Week	Topic	Lecture/Practical/Tutorial
1	Introduction to Physiology; scope and relevance to dentistry	Lecture
	Physiological terminology and levels of organization	Lecture
	Homeostasis and control systems	Tutorial
	Body fluid compartments and their regulation	Tutorial
	Parts and care of compound microscope	Practical
2	Functions of Cell and its organelles	Lecture
	Cell membrane structure and function	Lecture
	Intercellular junctions; Cell signalling	Tutorial
	Osmosis, Osmolarity, Osmotic pressure	Tutorial
	Osmotic fragility of red blood cells – principle and demonstration	Practical
3	Simple diffusion and facilitated diffusion	Lecture
	Active transport: Primary active transport	Lecture
	Active transport: Secondary active transport	Tutorial
	Bulk transport: Filtration and Vesicular transport	Tutorial
	Revision of membrane transport mechanisms	Tutorial
4	Structure and classification of neurons	Lecture
	Resting membrane potential	Lecture
	Action potential and its propagation	Tutorial
	Synapse: structure and function	Tutorial
	Introduction to PowerLab	Practical
5	Neuromuscular junction & Myasthenia gravis	Lecture
	Structure and properties of skeletal muscle	Lecture
	Excitation–contraction coupling	Tutorial
	Molecular mechanism of muscle contraction	Tutorial
	Recording of simple muscle twitch using PowerLab	Practical
6	Energetics of skeletal muscle; Oxygen debt.	Lecture
	Mechanics of skeletal muscle contraction	Lecture
	Motor unit and muscle tone	Tutorial
	Difference between skeletal, smooth and cardiac muscle	Tutorial
	Effect of repeated stimuli and fatigue in skeletal muscle	Practical
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7	Functions and composition of blood	Lecture
	Erythropoiesis and regulation	Lecture
	Hemoglobin synthesis, degradation and iron metabolism	Tutorial
	Classification of anemia and red cell indices	Tutorial
	ESR determination	Practical
8	Types of anemia	Lecture
	Polycythemia	Lecture
	Case studies on anemia	Tutorial
	Types of WBCs and their functions	Tutorial
	Hemoglobin estimation – Sahli's method	Practical
9	Immunity and its classification; Innate immunity	Lecture
	Cell mediated and humoral immunity	Lecture

	To determine differential leukocyte count	Practical
	Blood groups – ABO & Rh System	Tutorial
	Blood grouping and cross matching	Practical
10	Hemostasis – Vasoconstriction & Platelet plug formation	Lecture
	Hemostasis – Clot formation and its retraction	Lecture
	Hemostasis – Fibrinolytic mechanisms	Tutorial
	Disorders of hemostasis	Tutorial
	Bleeding time & Clotting time determination	Practical